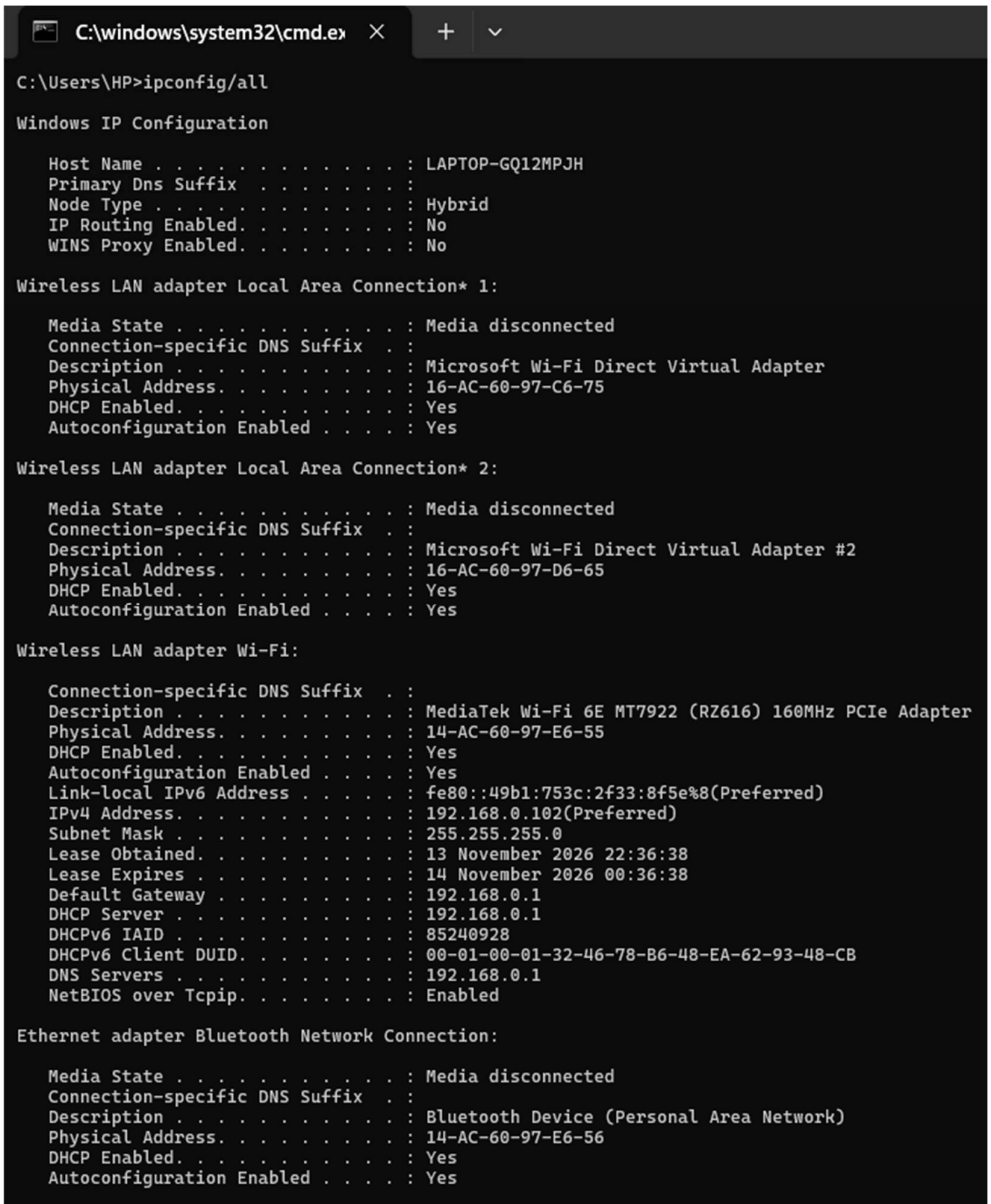
Mini Cybersecurity Challenge #1

DAY 08 — Networking & System Investigation Challenge

RUTUJA KADAM

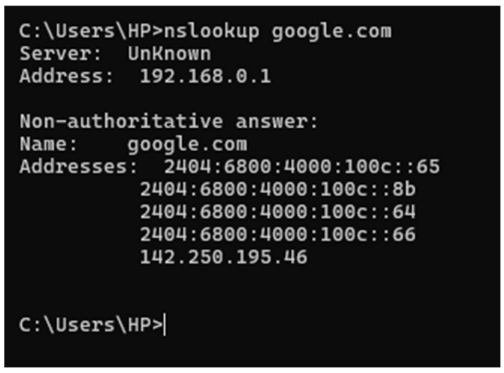
TASK 1 — NETWORK IDENTITY

Required field	Answer
IPv4 Address	192.168.0.102
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
DNS Server	192.168.0.1
MAC / Physical Address	14-AC-60-97-E6-55
DHCP Enabled	Yes



TASK 4 — DNS INVESTIGATION (15 POINTS)

Run: nslookup google.com



Assignment-

1. Which DNS server responded?

The DNS server responded from IP address 192.168.0.1. Its name is shown as Unknown.

2. Which IP address was returned?

The IPv4 address returned was 142.250.195.46.

IPv6 addresses were also returned:

- 2404:6800:4000:100c::65
- 2404:6800:4000:100c::8b
- 2404:6800:4000:100c::64
- 2404:6800:4000:100c::66

3. What is the purpose of DNS?

DNS (Domain Name System) translates human-readable domain names like google.com into IP addresses that computers use to communicate over a network.

TASK 5 — NETWORK PATH INVESTIGATION (15 POINTS)



TASK 5: ASSIGNMENT

1. How many visible hops appeared?

A total of **5 hops** appeared in the output. The 3rd hop showed Request timed out.

2. What is the first hop?

The first hop IP address is 192.168.0.1.

3. What device commonly represents the first hop?

The first hop commonly represents the **Default Gateway (Router)**.

4. What is traceroute/tracert used for?

Traceroute is used to identify the path taken by data packets from the source computer to the destination. It displays the intermediate hops (routers) and helps troubleshoot network connectivity issues.

Task 6 — Final Flag: Network Address Calculation

Flag : JVL{192_168_50_0}

